

Digital Twin Practice

Unplanned Risks Management

Strategies for
DISRUPTION
management (and
RIPPLE effect)



Co-funded by the
European Union



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Learning objectives

1. Develop analytical and management skills to analyze uncertainty in the supply chain.
2. We able to simulate uncertainty.
3. Two proactive strategies:
 - a) Increase inventory
 - b) Redundancy.



Use case: uncertainty in WHC production

- Reviewing the WHC use case
- How to create disruptive simulation scenarios



Disruption in WHC's supply chain !



Hello again!

A fire has destroyed the US distribution center and we are unable to make or accept deliveries for a month.



1) I need to estimate the impact of the disruption on supply chain performance for the following KPIs:

- Products received (inbound orders)
- Products delivered (outbound orders).

Lost sales

- Customer service level

2) I have to select the most efficient proactive and reactive strategies.

Proactive:

- Increasing inventory in the supply chain
- Have a back-up distribution center

Reactive:

- Rapid and costly recovery of the distribution center
- Slow and efficient recovery of the distribution center.



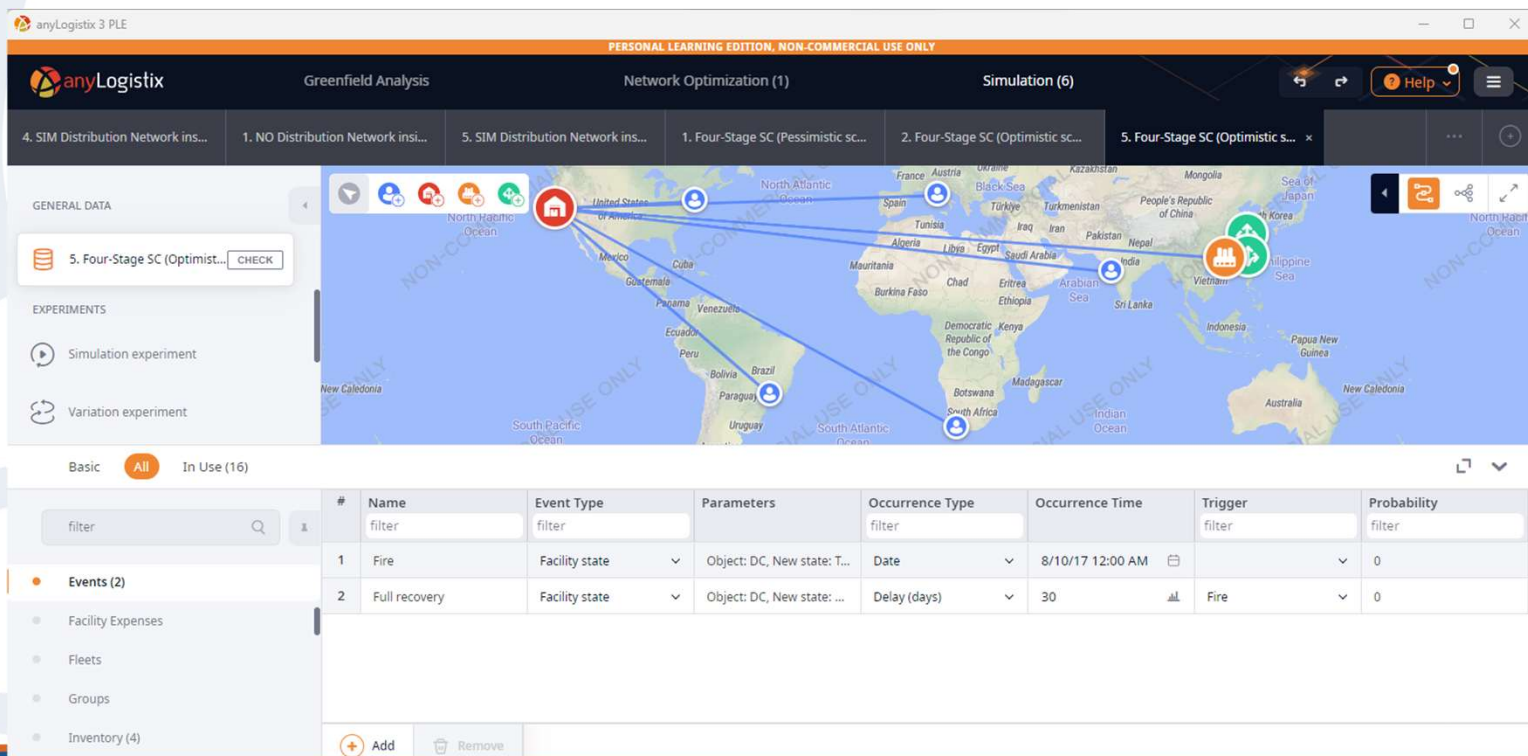
Supplier Taiwan, Supplier China



Factory

How to simulate a chain breakage?

- We use the event table that allows us to define different occurrences during the simulation.
- Typical events: chain breakage, recovery, etc...



The screenshot displays the anyLogistix 3 PLE software interface, which is used for simulating supply chain networks. The interface includes a map of the world with various countries and regions labeled, and a network diagram showing connections between different locations. The software is running a simulation titled "Simulation (6)".

The interface is divided into several sections:

- GENERAL DATA:** Includes a section for "5. Four-Stage SC (Optimistic...)" with a "CHECK" button.
- EXPERIMENTS:** Includes a section for "Simulation experiment" and "Variation experiment".
- Events Table:** A table listing events, including "Fire" and "Full recovery".

#	Name	Event Type	Parameters	Occurrence Type	Occurrence Time	Trigger	Probability
1	Fire	Facility state	Object: DC, New state: T...	Date	8/10/17 12:00 AM		0
2	Full recovery	Facility state	Object: DC, New state: ...	Delay (days)	30	Fire	0



The events

- They can only be used in simulation-type experiments:
 - Simulation, Variation, Comparison, Safety Stock Estimation, and Risk Analysis.
- Occurrence of events:
 - Simple events, based on concrete planning.
 - Chained" and "triggered" events based on other events and their sequence.
- a) Events planned at a certain point in time (date/time).
- b) Events by "trigger", from a certain time or event.

Types of events

- Types of events: depend on the object they affect.

SITE (DC, Fabrica, ...)	costS	DEMAND	ROUTES	PRODUCTION LINE	FLEETS
- Change of status - Activate - Temporarily closed - Inactive	- change in its structure or value - By quantity - Per site.	- Changes in its structure or value - By quantity - By customer	- Change of status - Activate - Temporarily closed - Inactive	- Change of status - By factory - By product - By state: - Operative - Inoperative, receives orders - Hard" operative: no orders can be received - Closing	- Fleet resizing - P.e. Breakdowns or maintenance - Per site - By type of vehicle - By volume

- Inoperative objects appear in GRAY on the map.



Exercise 6, 7: WHC scenarios baseline and disruption

- Baseline experiment
- Disruptive experiment



WHC's supply chain baseline



We need to analyze supply chain disruption in terms of:

- Incoming orders (product orders)
- Orders delivered
- Lost sales
- Customer service level
- Impact on financial indicators

Customers (5)

DCs and Factories (2)

Demand (5)

Inventory (4)

Locations (9)

Paths (4)

2	Brazil	Customer	Brazil location	Include	Additional par
3	South Africa	Customer	South Africa location	Include	Additional par
4	Italy	Customer	Italy location	Include	Additional par
5	India	Customer	India location	Include	Additional par
	Add		Remove		Generate



Supplier Taiwan, Supplier China



Factory



DC



(All customers)

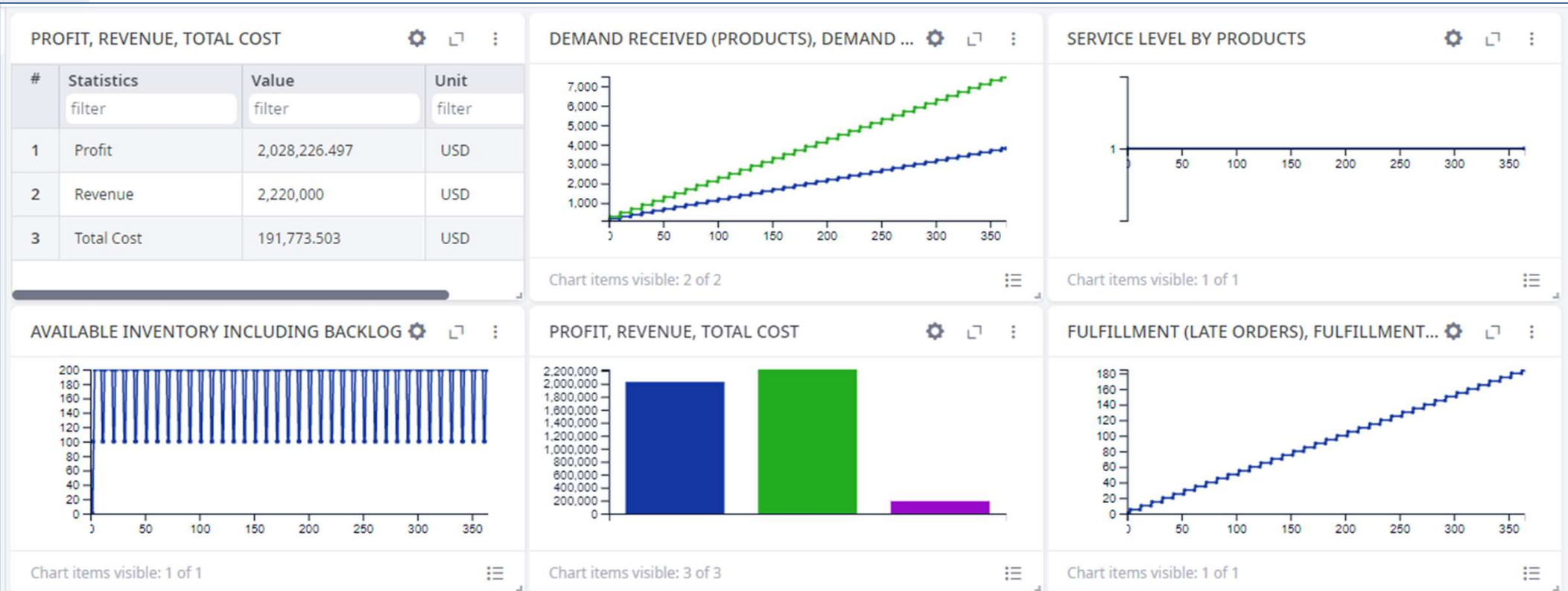


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Exercise 6: WHC Baseline Simulation without SC disruption

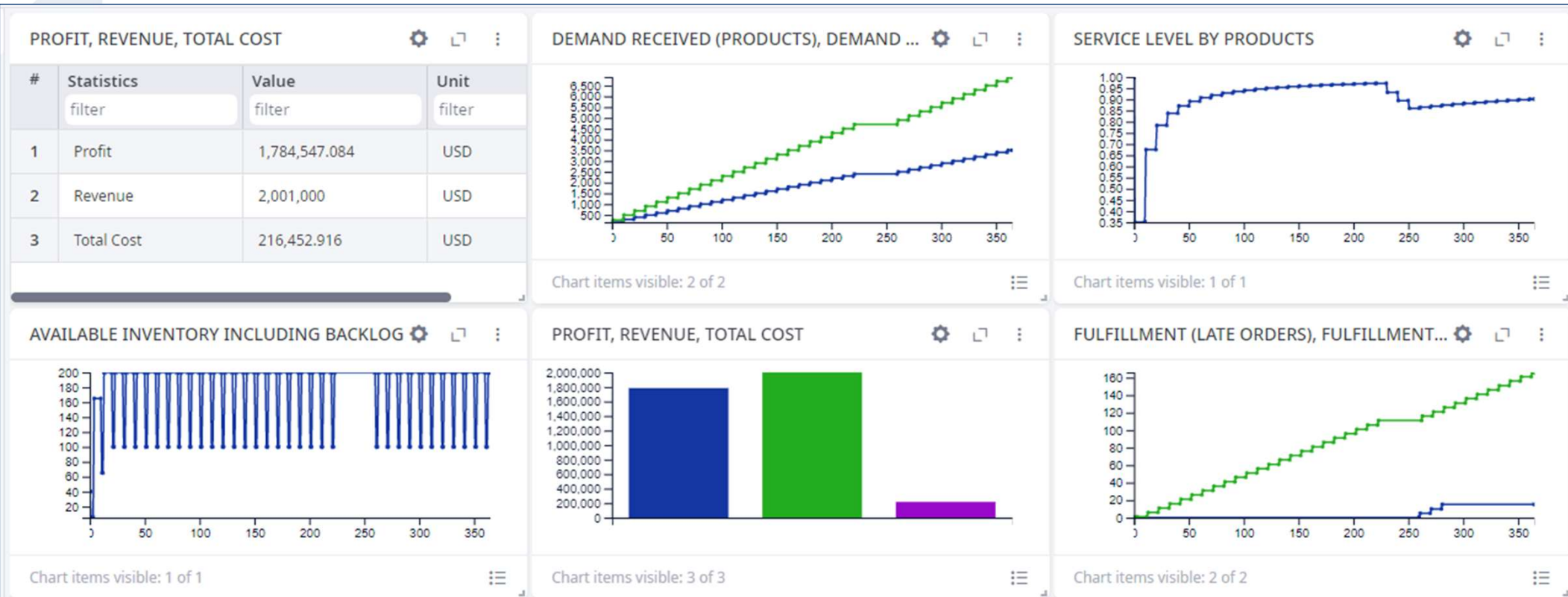
- We set the probability of occurrence to "0" (will not occur).





Exercise 7: WHC Disruption: the fire occurs

- We set the probability of occurrence to "1" (it will necessarily occur).





Scenario comparison: financial

- We can see that a 30-day supply chain disruption has a severe economic impact, not only on sales but also on costs.

PROFIT, REVENUE, TOTAL COST			
#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	2,028,226.497	USD
2	Revenue	2,220,000	USD
3	Total Cost	191,773.503	USD

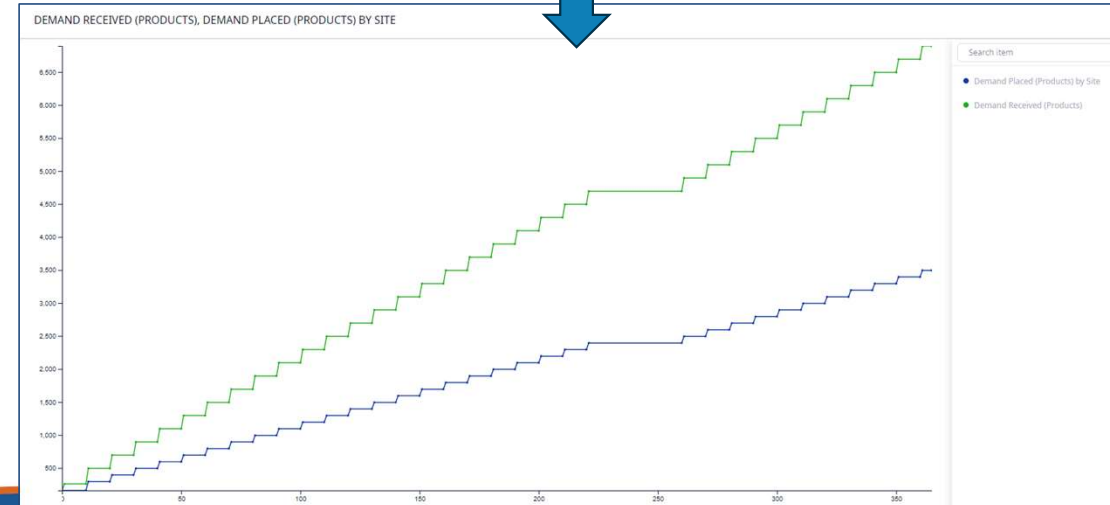
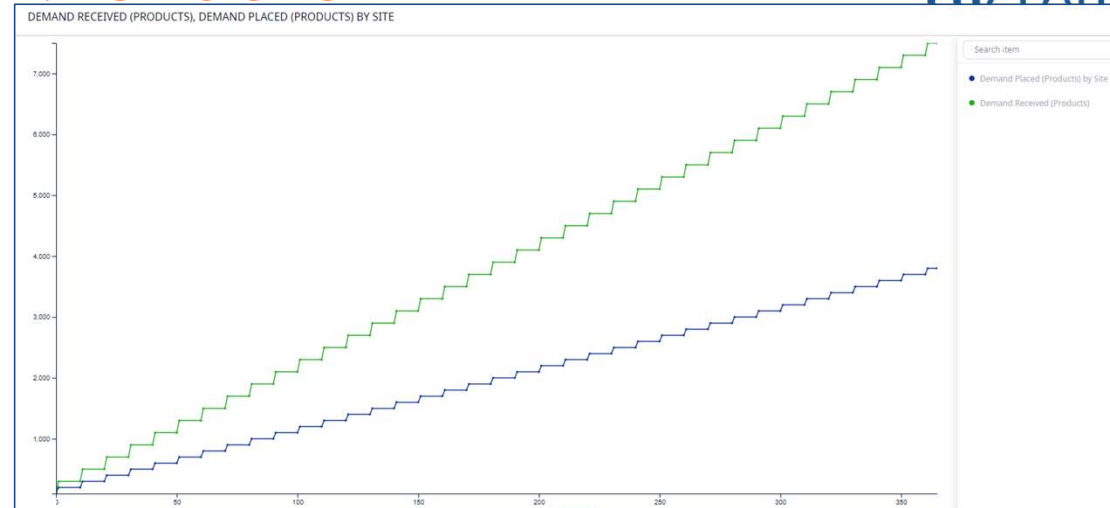


PROFIT, REVENUE, TOTAL COST			
#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,784,547.084	USD
2	Revenue	2,001,000	USD
3	Total Cost	216,452.916	USD



Scenario comparison: Orders

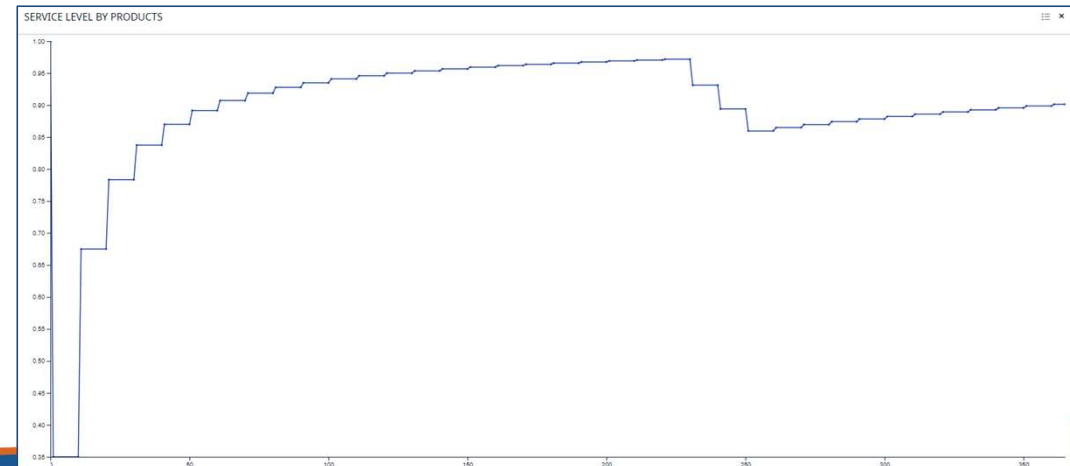
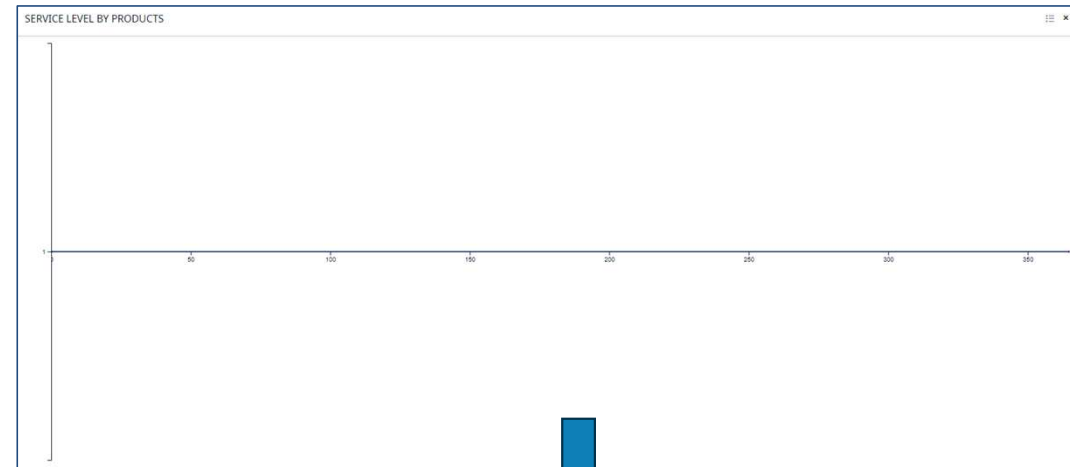
- We note how for 30 days we are unable to serve orders once safety inventories are consumed.





Scenario comparison: service level

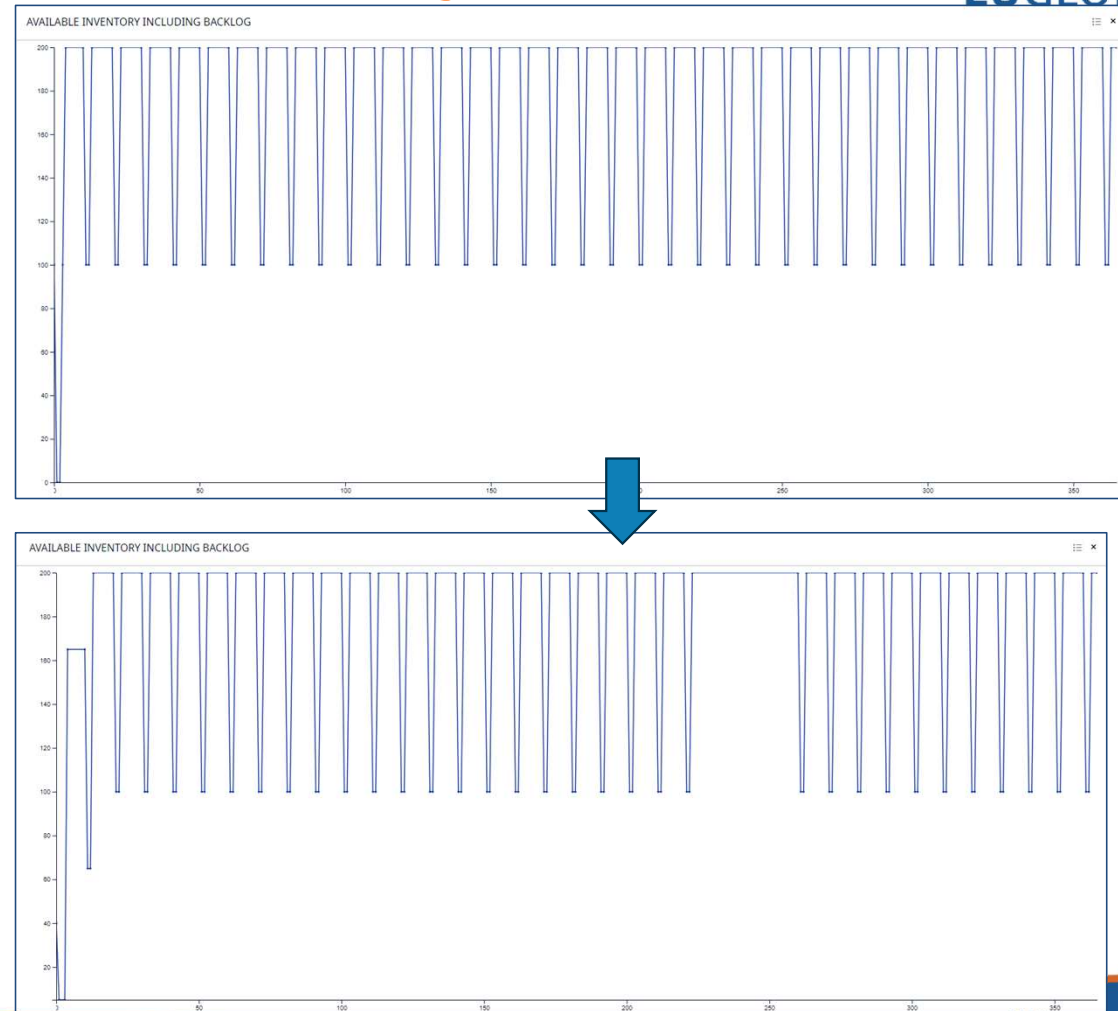
- We also see how service levels are affected.
- The safety stock mitigates the problem somewhat, but it is difficult to recover from it because of the orders accumulated over time and the production capacity (which is limited).





Scenario comparison: inventory

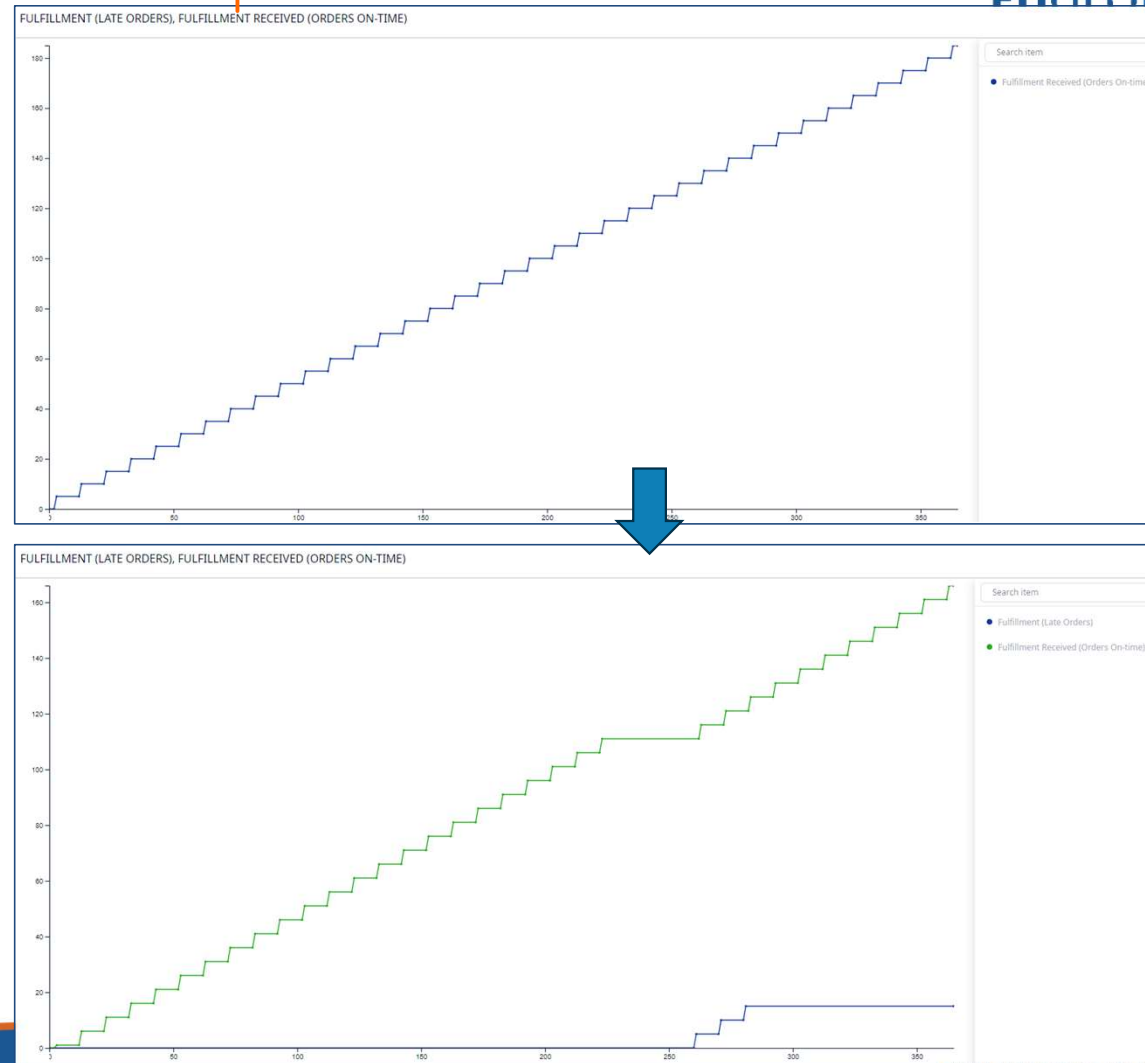
- We see how the order backlog gets stuck in the distribution center.





Scenario comparison: replenishment

- Once the disruption is resolved, the chain is restarted.





Prevention vs Reaction vs cost



The supply chain manager must select the most efficient proactive and reactive strategies.

- Proactive strategies:
 - Increase inventory in the supply chain.
 - Create a backup distribution center.
- Reactive strategies:
 - A. Distribution Center Rapid Recovery Plan → cost problem.
 - B. Slow recovery plan for the distribution center → more cost effective.



Supplier Taiwan, Supplier China



Factory



DC



(All customers)



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Exercise 8: Increase inventory capacity to protect against disruption



Proactive strategies: protection against disruption by increasing inventory

- As we have seen in the example, the inventory acts as a "cushion" to protect the SC disruption.
- We modify the volume of the min-Max (s,S):
 - (100,200) → (100,400)
- We do not touch storage costs (although we should).
- And we simulate again...



Scenario comparison: financial

- Failure to sell more....
- On the other hand, we have much higher costs, which is a drag on our profit...

PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	2,028,226.497	USD
2	Revenue	2,220,000	USD
3	Total Cost	191,773.503	USD



PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,784,547.084	USD
2	Revenue	2,001,000	USD
3	Total Cost	216,452.916	USD



PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,754,525.203	USD
2	Revenue	2,001,000	USD
3	Total Cost	246,474.797	USD



It is necessary to analyze the costs...

Average Cost per Item	x
Average Cost per Order	x
Average Number of Vehicles Used	x
Inventory Spend	x
Production Cost	x
Total Cost	x
Transportation Cost	x
Vehicle Capacity Utilization	x

AVERAGE COST PER ITEM, AVERAGE COST PER ORDER, INVENTORY SPEND, INV...

#	Statistics	Value	Unit
	filter	filter	filter
1	Average Cost per Order	1,061.044	USD
2	Average Cost per Item	31.393	USD
3	Inventory Spend	40,000	USD
4	Production Cost	175,750	USD
5	Total Cost	216,452.916	USD
6	Vehicle Capacity Utilization	0.001	Ratio
7	Transportation Cost	702.916	USD
8	Average Number of Vehicles ...	0.586	Vehicle



AVERAGE COST PER ITEM, AVERAGE COST PER ORDER, AVERAGE NUMBER OF V...

#	Statistics	Value	Unit
	filter	filter	filter
1	Average Cost per Order	1,354.257	USD
2	Average Cost per Item	34.739	USD
3	Inventory Spend	60,000	USD
4	Production Cost	185,750	USD
5	Total Cost	246,474.797	USD
6	Vehicle Capacity Utilization	0.001	Ratio
7	Transportation Cost	724.797	USD
8	Average Number of Vehicles Used	0.501	Vehicle

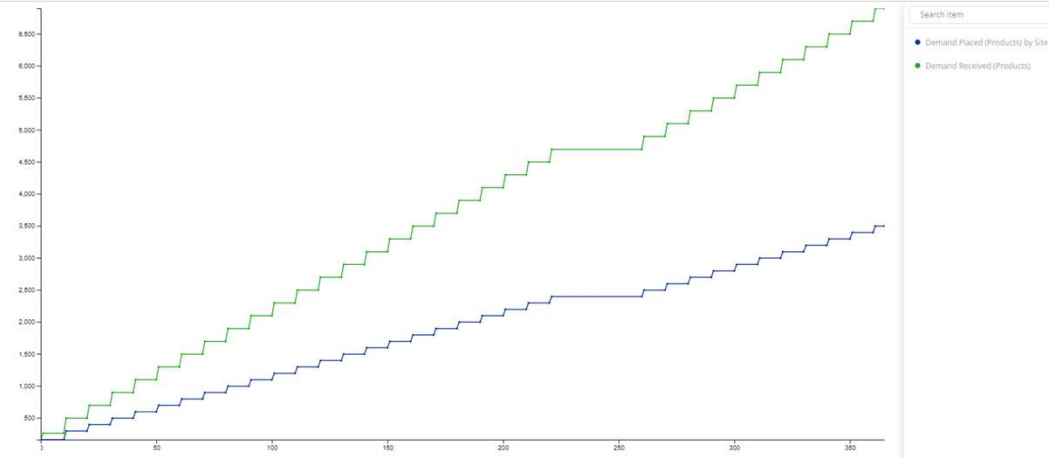


Scenario comparison: Orders

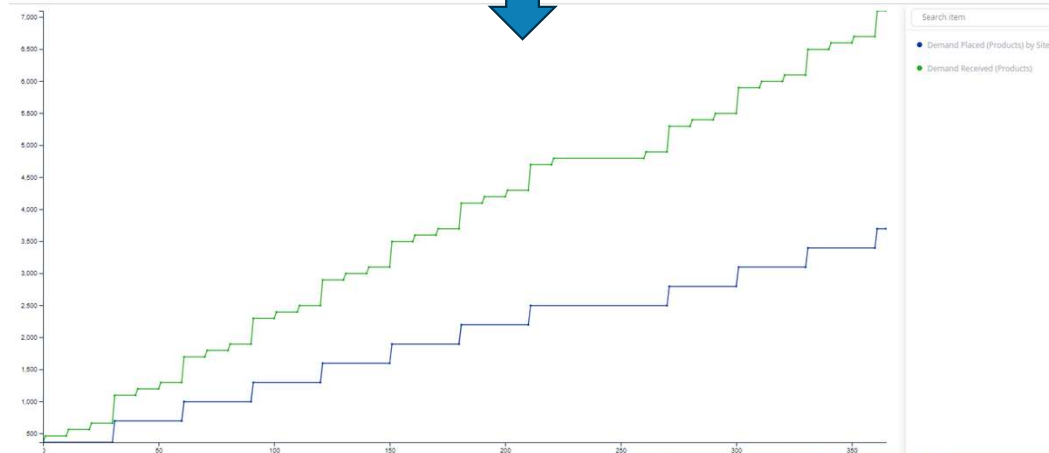
- Nothing has changed?
- Yes, we have delivered fewer orders



DEMAND RECEIVED (PRODUCTS), DEMAND PLACED (PRODUCTS) BY SITE



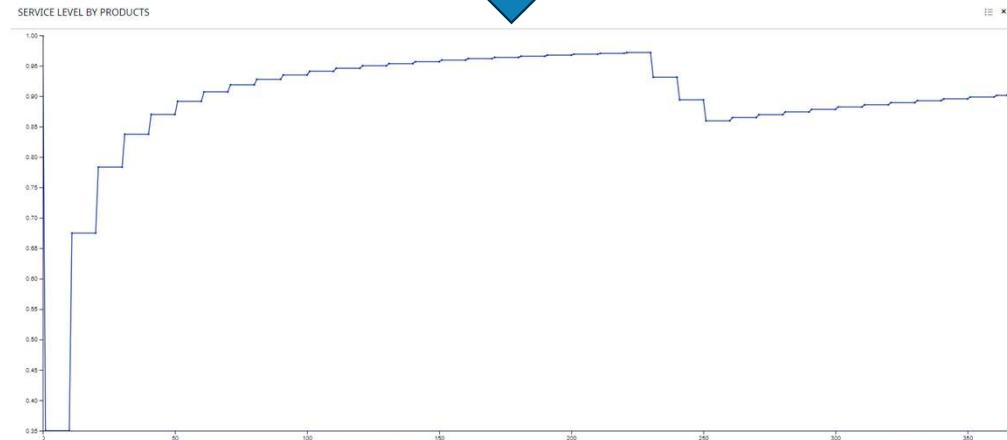
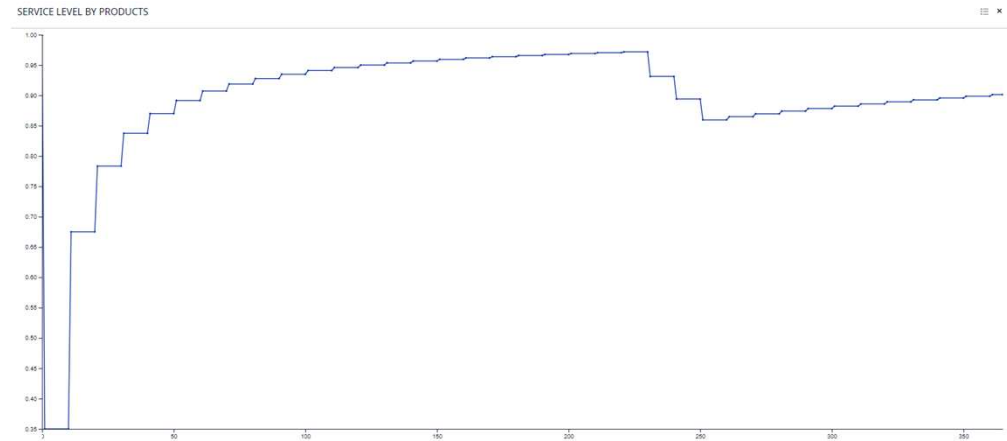
DEMAND RECEIVED (PRODUCTS), DEMAND PLACED (PRODUCTS) BY SITE





Scenario comparison: service level

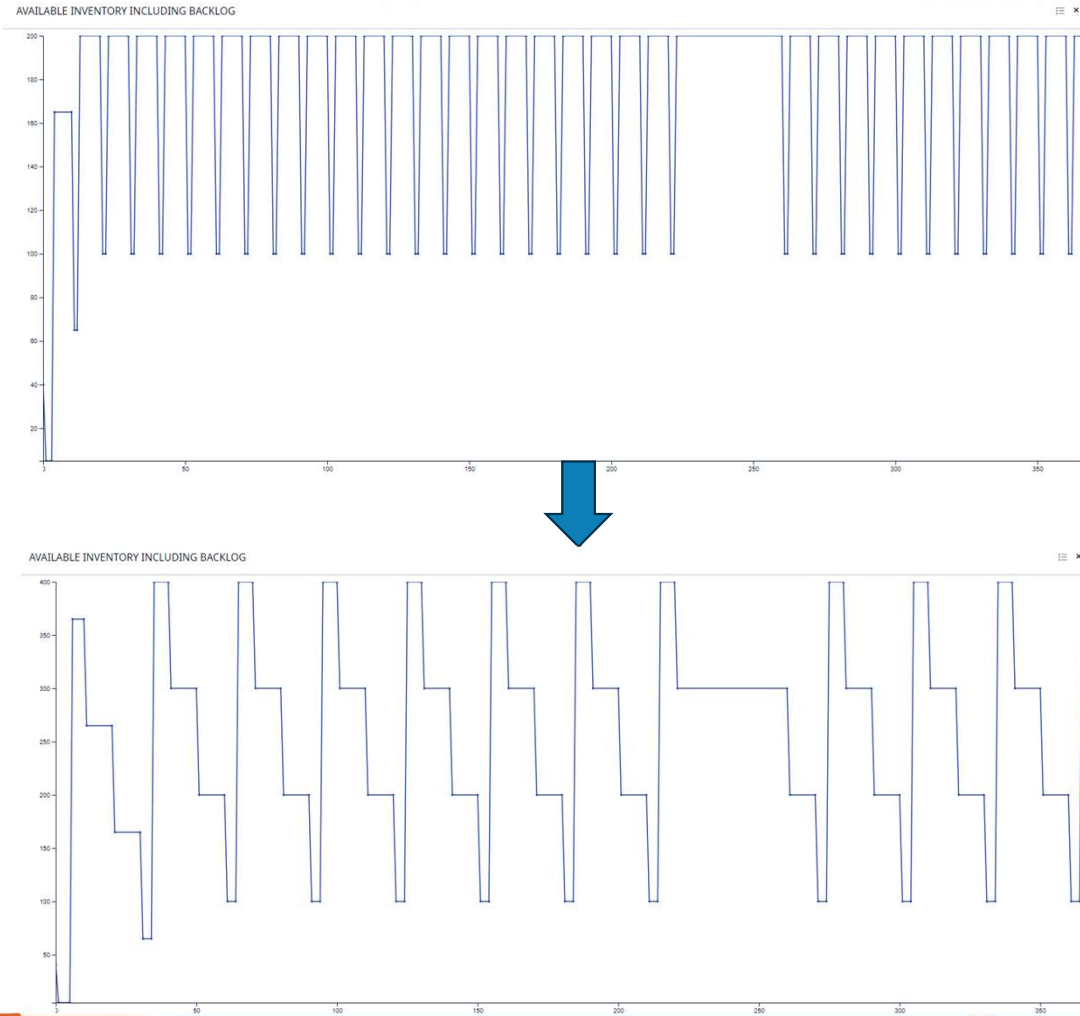
- And we continue to maintain a poor level of service below expectations.





Scenario comparison: inventory

- And we have large inventory fluctuations.





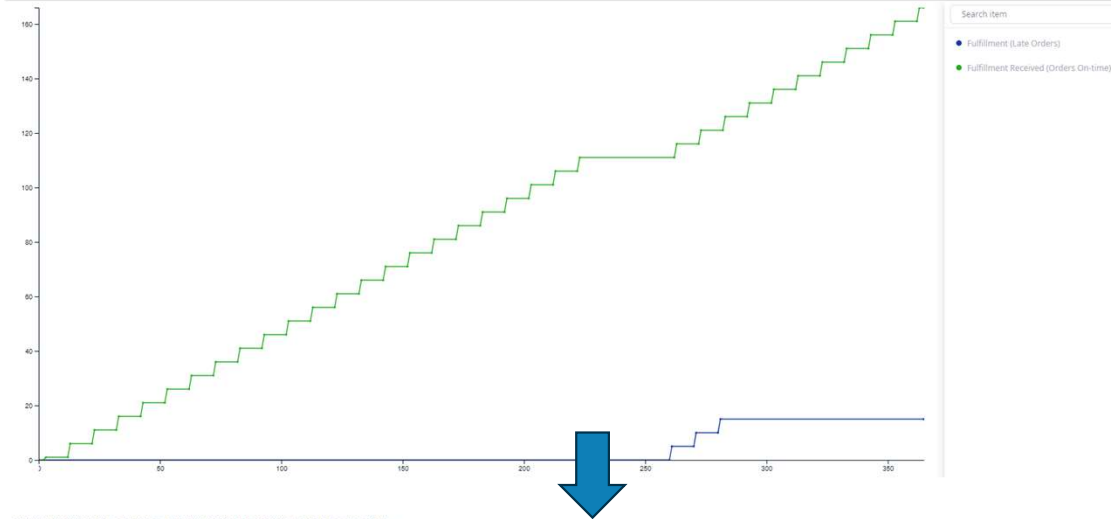
Scenario comparison: replenishment

- Where we continue to maintain the warehouse replacement rate.

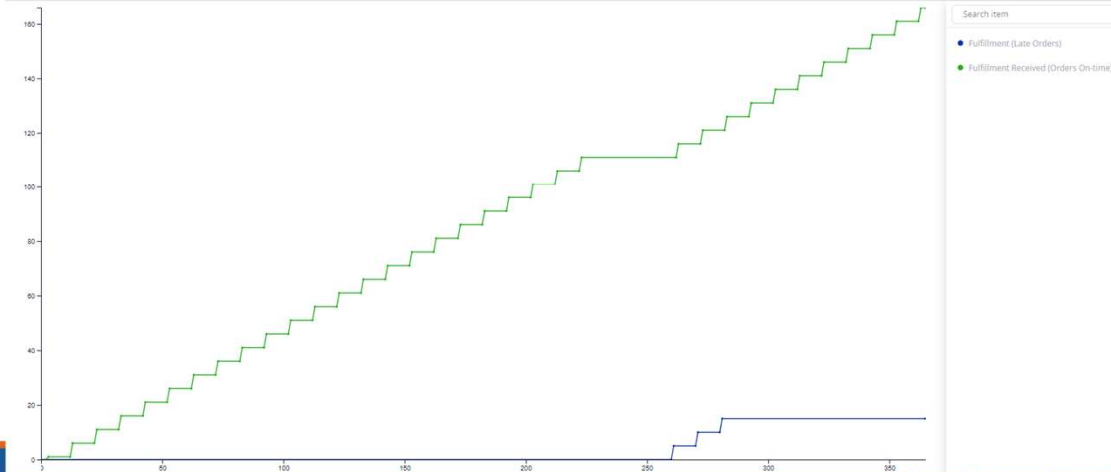
We have just seen that increasing our inventory policies has not solved our problem...

→ In this case we are limited by our production capacity !!!!

FULFILLMENT (LATE ORDERS), FULFILLMENT RECEIVED (ORDERS ON-TIME)



FULFILLMENT (LATE ORDERS), FULFILLMENT RECEIVED (ORDERS ON-TIME)





Exercise 9: redundancy to protect against disruption





Proactive strategies: implementing a backup distribution center

- This DC will not be part of our "normal" SC, but we can activate it in case of need.
- How will we activate it? → Through events

#	Name filter	Event Type filter	Parameters	Occurrence Type filter	Occurrence Time	Trigger filter	Probability filter
1	Fire	Facility state ▼	Object: DC, New state: Te...	Date ▼	8/10/17 12:00 AM 📅	▼	1
2	Full recovery	Facility state ▼	Object: DC, New state: O...	Delay (days) ▼	30 📅	Fire ▼	1
3	In back-up DC	Facility state ▼	Object: Back-up DC, New ...	Date ▼	8/10/17 12:00 AM 📅	Fire ▼	1
4	Out back-up DC	Facility state ▼	Object: Back-up DC, New ...	Date ▼	9/10/17 12:00 AM 📅	In back-up DC ▼	1

- "In case of receiving the 'Fire' event on the normal DC, change the DC-Backup status to 'Active'."
- "When the 'Fire' event ceases, change the DC-Backup status to 'Inactive'."



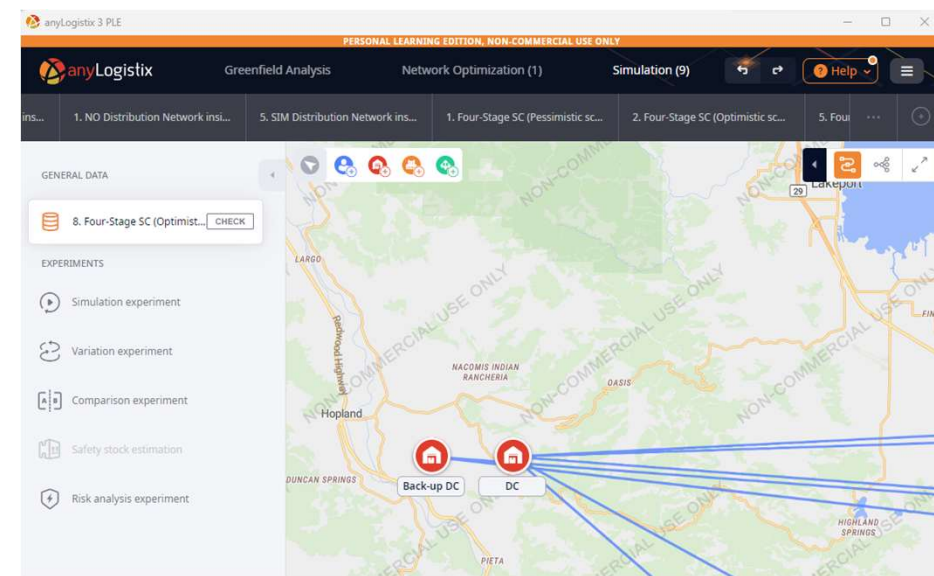
Required configuration

#	Name	Type	Location	Initially Open	Inclusion Type	Capacity	Capacity Unit	Aggregate Orders by L	Additional Parameters	Icon
	filter	filter	filter	filter	filter	filter	filter	filter		
1	DC	DC	DC location	☒	Include	0	m ³	☐	Additional parameters	
2	Factory	Factory	Factory location	☒	Include	0	m ³	☐	Additional parameters	
3	Back-up DC	DC	Back-up DC location	☒	Include	0	m ³	☐	Additional parameters	

- We create the new DC-Backup center in the same location (approx.) as the normal one.

- We add your costs:

#	Facility	Expense Type	Value	Currency
	filter	filter	filter	filter
1	Back-up DC	Initial cost	40,000	USD



Required configuration

- We add the necessary inventory policies:

anyLogistix

Greenfield Analysis Network Optimization (1) Simulation (9)

1. NO Distribution Network insi... 5. SIM Distribution Network ins... 1. Four-Stage SC (Pessimistic sc... 2. Four-Stage SC (Optimistic sc... 5. Four-Stage SC (Optimistic sc... 6. Four-Stage SC (Optimistic sc... 7. Four-Stage SC (Optimistic sc... 8. Four-Stage SC (Optimistic sc...

GENERAL DATA

8. Four-Stage SC (Optimistic... CHECK

EXPERIMENTS

Simulation experiment 1

Variation experiment

Comparison experiment

Safety stock estimation

Basic All In Use (17)

#	Facility	Product	Policy Type	Policy Parameters	Initial Stock, units	Periodic Check	Period	First Periodic Check	Policy Basis	Stock C
1	DC	Smartphone	Inventory Policy Mir	s=100, S=200	150	☐	1	1/1/17 12:00 AM	Quantity	0
2	Factory	Smartphone	Inventory Policy Mir	s=30, S=60	40	☐	1	1/1/17 12:00 AM	Quantity	0
3	Factory	Chip	Unlimited Inventory	Unlimited	∞	☐	1	1/1/17 12:00 AM	Quantity	0
4	Factory	Display	Unlimited Inventory	Unlimited	∞	☐	1	1/1/17 12:00 AM	Quantity	0
5	Back-up DC	Smartphone	Inventory Policy Mir	s=100, S=200	50	☐	1	1/1/17 12:00 AM	Quantity	0

Required configuration

- We add the necessary routes:

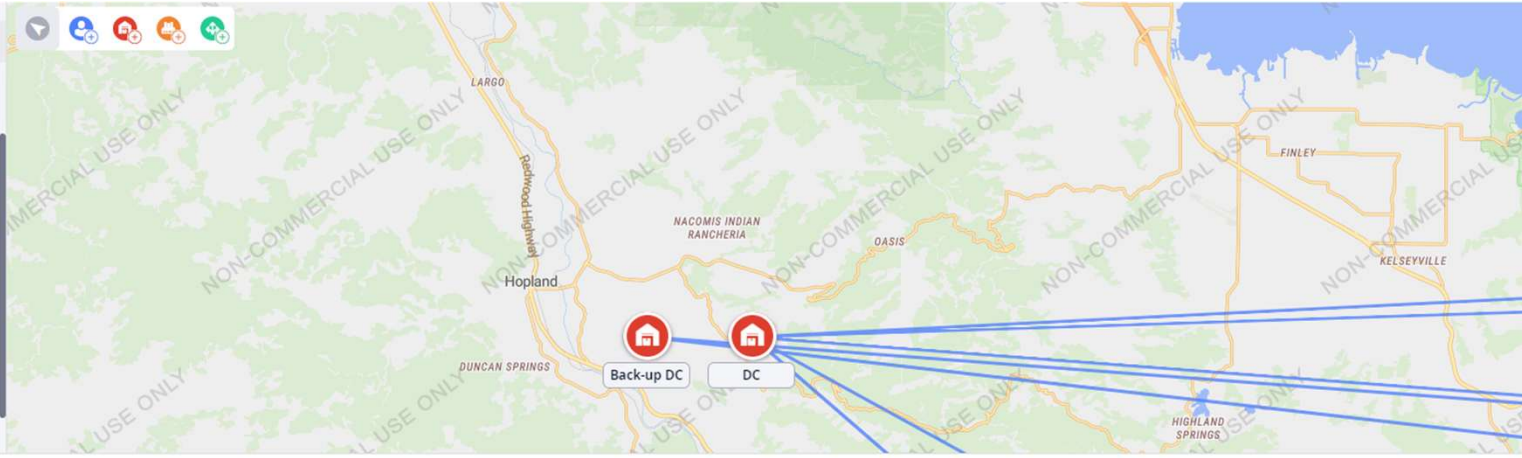
GENERAL DATA

8. Four-Stage SC (Optimist... CHECK)

EXPERIMENTS

- Simulation experiment
- Variation experiment
- Comparison experiment
- Safety stock estimation

Basic All In Use (17)



#	From	To	Cost Calculation	Cost Calculation Par	CO2 Calculation Par	Currency	Distance	Distance Unit	Transportation Time
1	Supplier China locat	Factory location	Distance-based with	0.5 * distance + 0 pe...	0 * distance + 0 per ...	USD	0	km	0
2	Supplier Taiwan loc	Factory location	Distance-based with	0.8 * distance + 0 pe...	0 * distance + 0 per ...	USD	0	km	0
3	Factory location	DC location	Product&distance-b	0.01 * product (m³) *...	0 * product (m³) * di...	USD	0	km	2
4	DC location	(All locations)	Product&distance-b	0.01 * product (m³) *...	0 * product (m³) * di...	USD	0	km	2
5	Back-up DC location	(All locations)	Product&distance-b	0.01 * product (m³) *...	0 * product (m³) * di...	USD	0	km	2
6	Factory location	Back-up DC location	Product&distance-b	0.01 * product (m³) *...	0 * product (m³) * di...	USD	0	km	2

Required configuration

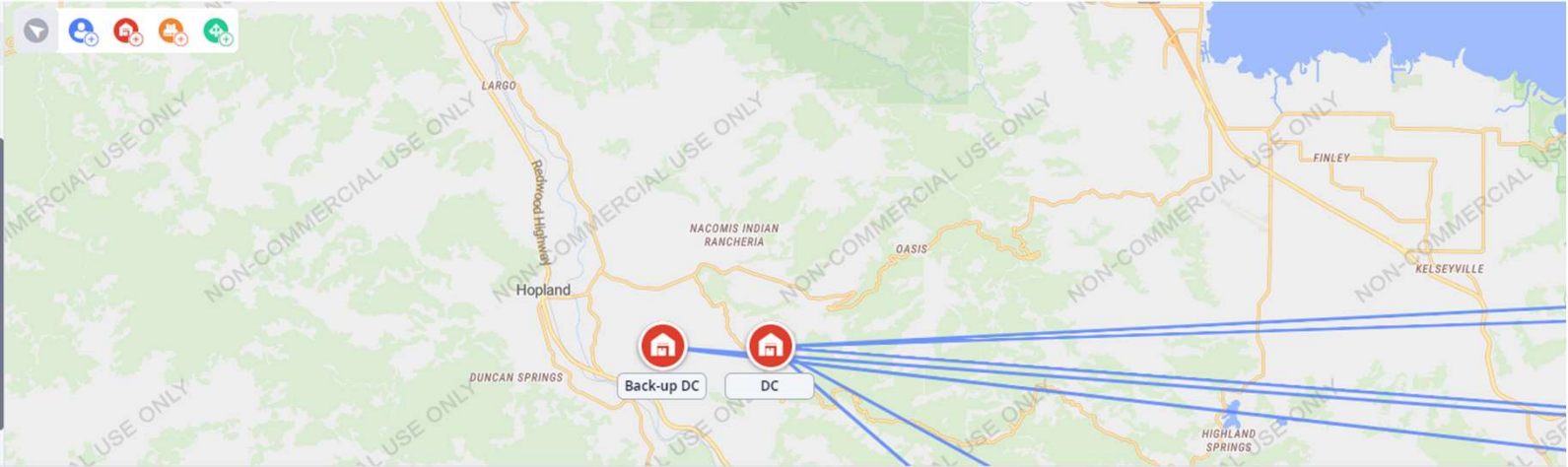
- We add shipping policies:

GENERAL DATA

8. Four-Stage SC (Optimist... ☐ CHECK

EXPERIMENTS

- Simulation experiment
- Variation experiment
- Comparison experiment
- Safety stock estimation



Basic All **In Use (17)**

#	Sources	Destinations	Product	Vehicle Type	Type	Parameters	Priority	Days of Week	Start Time
1	Supplier China	Factory	(All products)	Truck	LTL	Partial delivery	FIFO		12:00 AM
2	Supplier Taiwan	Factory	(All products)	Ferry	LTL	Partial delivery	FIFO		12:00 AM
3	Factory	DC	(All products)	Airplane	LTL	Partial delivery	FIFO		12:00 AM
4	DC	Selected 2	(All products)	Airplane	LTL	Partial delivery	FIFO		12:00 AM
5	Back-up DC	Selected 2	(All products)	Airplane	LTL	Partial delivery	FIFO		12:00 AM
6	Factory	Back-up DC	(All products)	Airplane	LTL	Partial delivery	FIFO		12:00 AM

Required configuration

- We add sourcing policies:

GENERAL DATA

8. Four-Stage SC (Optimist... ☐ CHECK

EXPERIMENTS

- Simulation experiment
- Variation experiment
- Comparison experiment
- Safety stock estimation

Basic All In Use (17)

filter

#	Delivery Destination	Sources	Product	Type	Parameters
1	Factory	Supplier China	Display	Closest (Dynamic Sources)	No parameters
2	Factory	Supplier Taiwan	Chip	Closest (Dynamic Sources)	No parameters
3	DC	Factory	Smartphone	Closest (Dynamic Sources)	No parameters
4	(All customers)	Selected 2	Smartphone	Closest (Dynamic Sources)	No parameters
5	Back-up DC	Factory	Smartphone	Closest (Fixed Source)	No parameters

Paths (6)

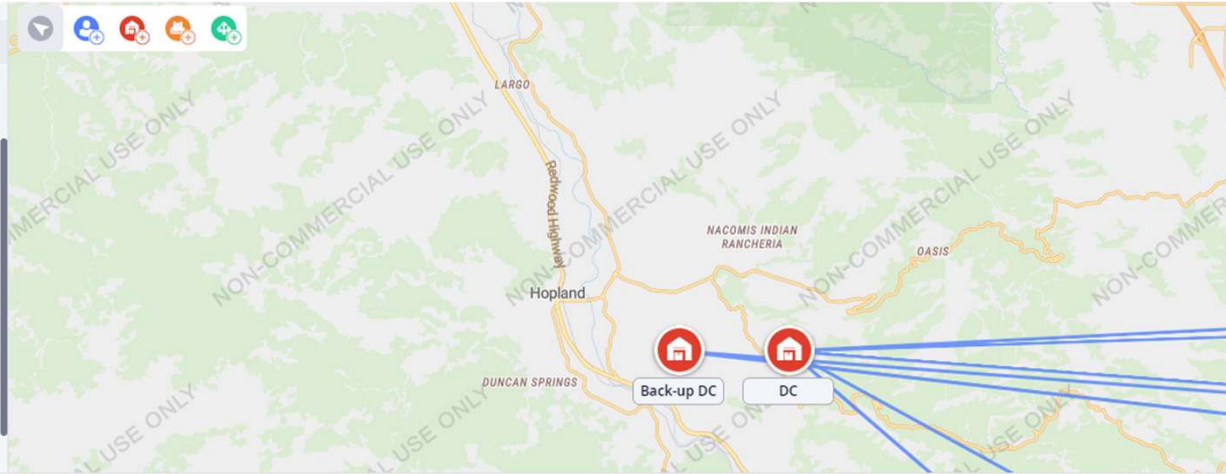
Periods (1)

Production (1)

Products (3)

Shipping (6)

Sourcing (5)





And we simulate and compare: financial

PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	2,028,226.497	USD
2	Revenue	2,220,000	USD
3	Total Cost	191,773.503	USD



PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
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1	Profit	1,784,547.084	USD
2	Revenue	2,001,000	USD
3	Total Cost	216,452.916	USD



PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,754,525.203	USD
2	Revenue	2,001,000	USD
3	Total Cost	246,474.797	USD



PROFIT, REVENUE, TOTAL COST

#	Statistics	Value	Unit
	filter	filter	filter
1	Profit	1,983,215.583	USD
2	Revenue	2,220,000	USD
3	Total Cost	236,784.417	USD

- This is something else: we managed to recover our original level of sales, even though we have a higher cost due to the activation of the new DC-backup.



Delving deeper into costs

AVERAGE COST PER ITEM, AVERAGE COST PER ORDER, INVENTORY SPEND, INV...

#	Statistics	Value	Unit
	filter	filter	filter
1	Average Cost per Order	1,061.044	USD
2	Average Cost per Item	31.393	USD
3	Inventory Spend	40,000	USD
4	Production Cost	175,750	USD
5	Total Cost	216,452.916	USD
6	Vehicle Capacity Utilization	0.001	Ratio
7	Transportation Cost	702.916	USD
8	Average Number of Vehicles ...	0.586	Vehicle



AVERAGE COST PER ITEM, AVERAGE COST PER ORDER, AVERAGE NUMBER OF V...

#	Statistics	Value	Unit
	filter	filter	filter
1	Average Cost per Order	1,354.257	USD
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6	Vehicle Capacity Utilization	0.001	Ratio
7	Transportation Cost	724.797	USD
8	Average Number of Vehicles Used	0.501	Vehicle



AVERAGE COST PER ITEM, AVERAGE COST PER ...

#	Statistics	Value	Unit
	filter	filter	filter
1	Average Cost per Or...	1,091.172	USD
2	Average Cost per Item	31.156	USD
3	Inventory Spend	40,000	USD
4	Production Cost	196,000	USD
5	Total Cost	236,784.417	USD
6	Vehicle Capacity Utili...	0.001	Ratio
7	Transportation Cost	784.417	USD
8	Average Number of ...	0.646	Vehicle

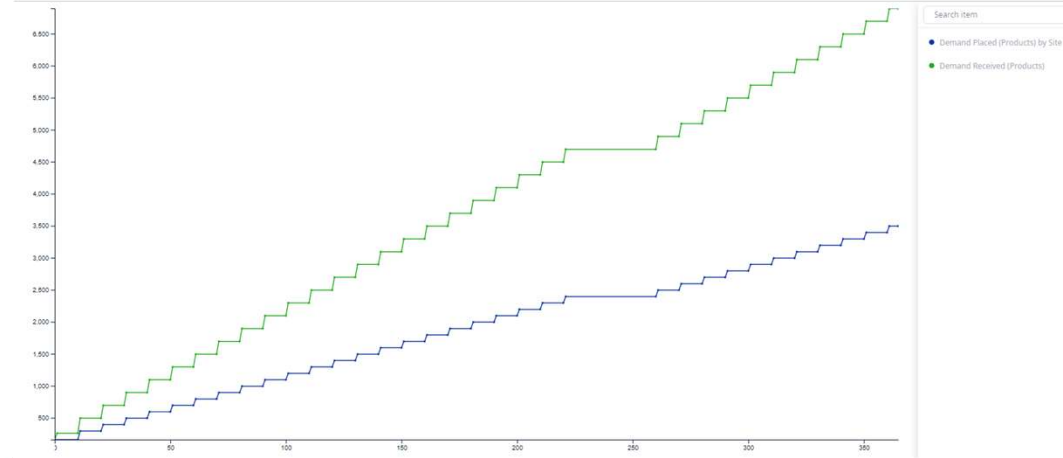


Scenario comparison: Orders

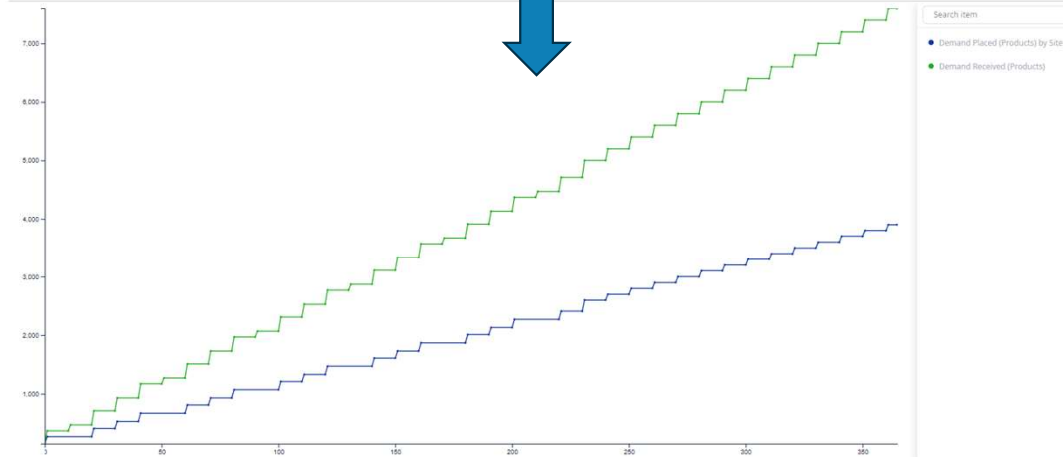
- We are now able to fill all orders on time.



DEMAND RECEIVED (PRODUCTS), DEMAND PLACED (PRODUCTS) BY SITE



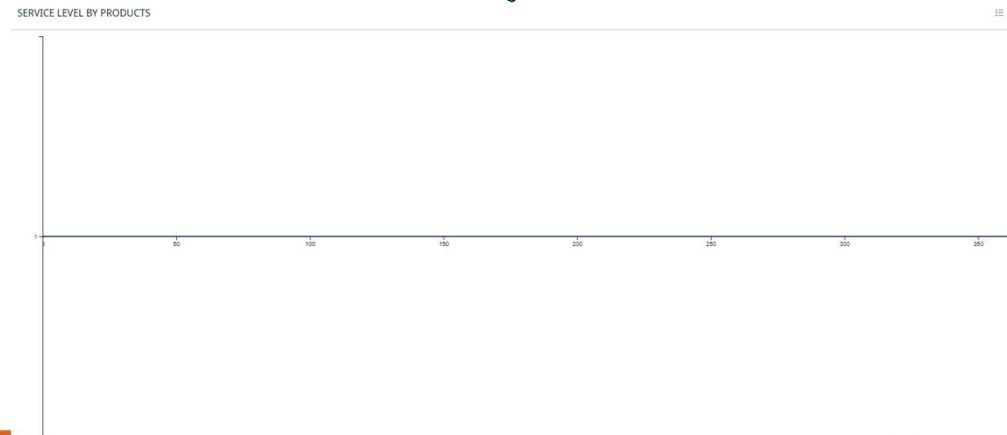
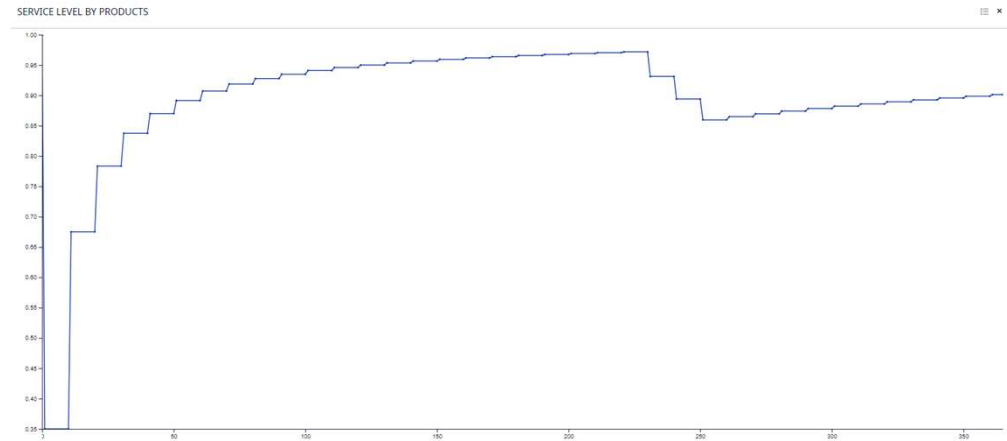
DEMAND RECEIVED (PRODUCTS), DEMAND PLACED (PRODUCTS) BY SITE





Scenario comparison: service level

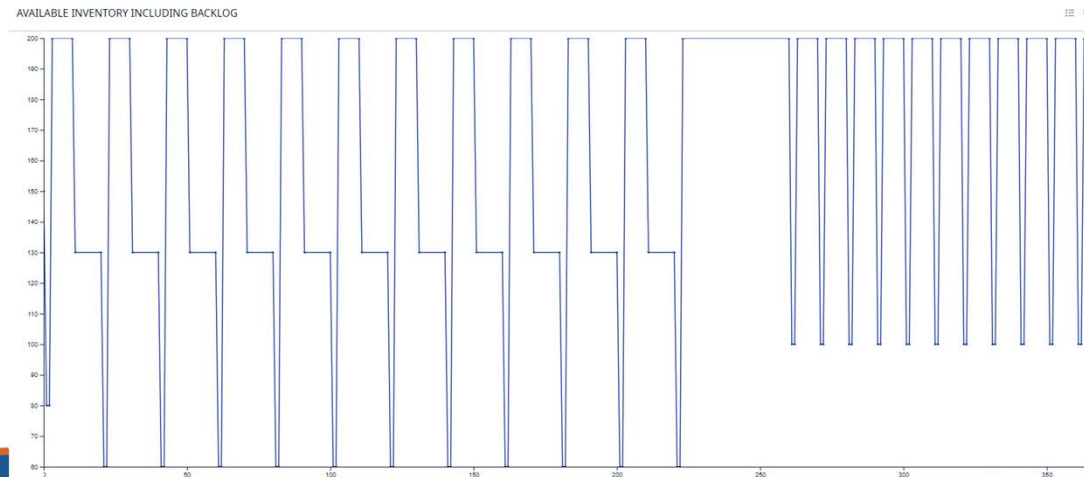
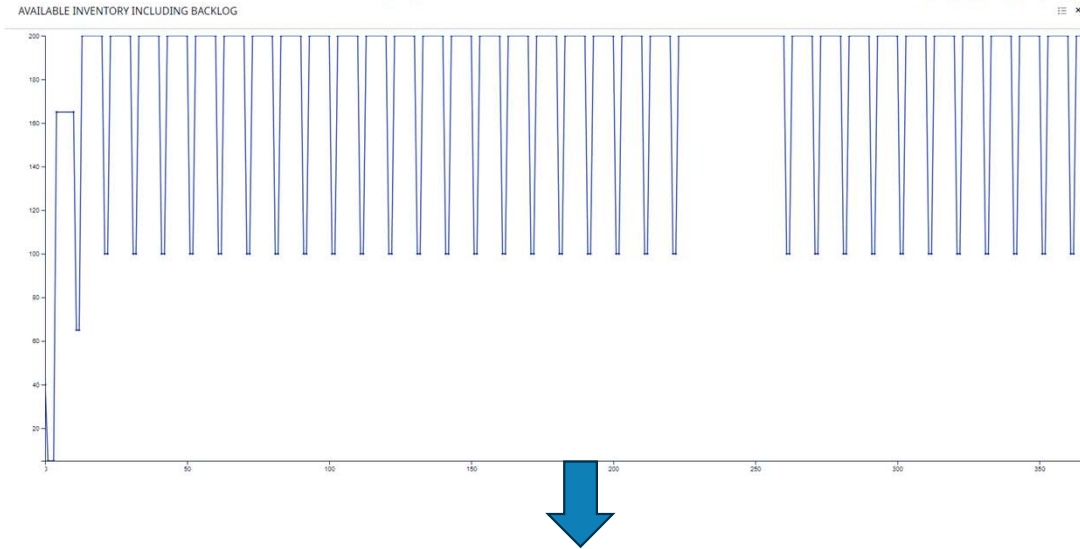
- ... which can be seen in the level of service we provide to our customers.





Scenario comparison: inventory

- However, what we see is that the inventory+order backlog acts as a backup to our SC.

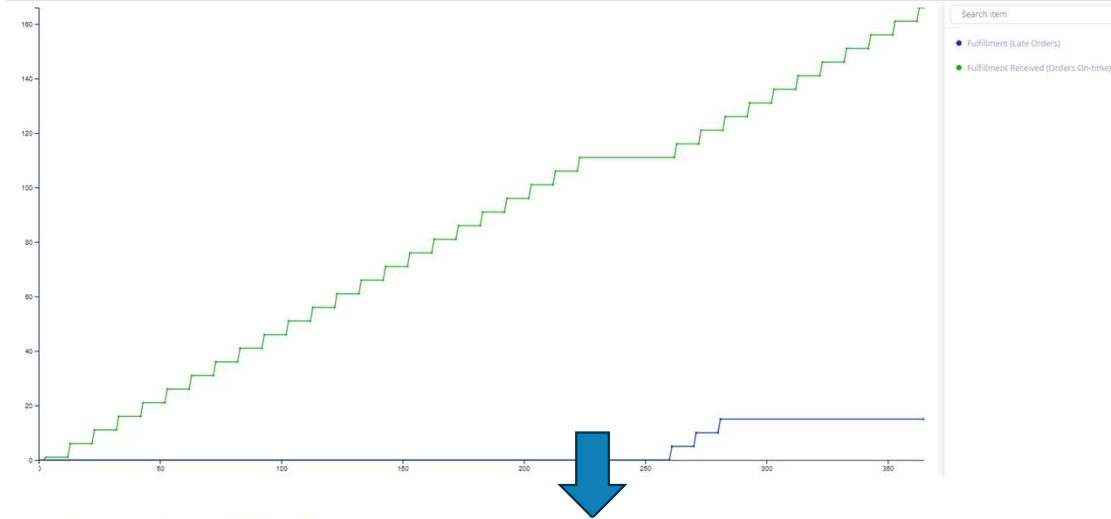




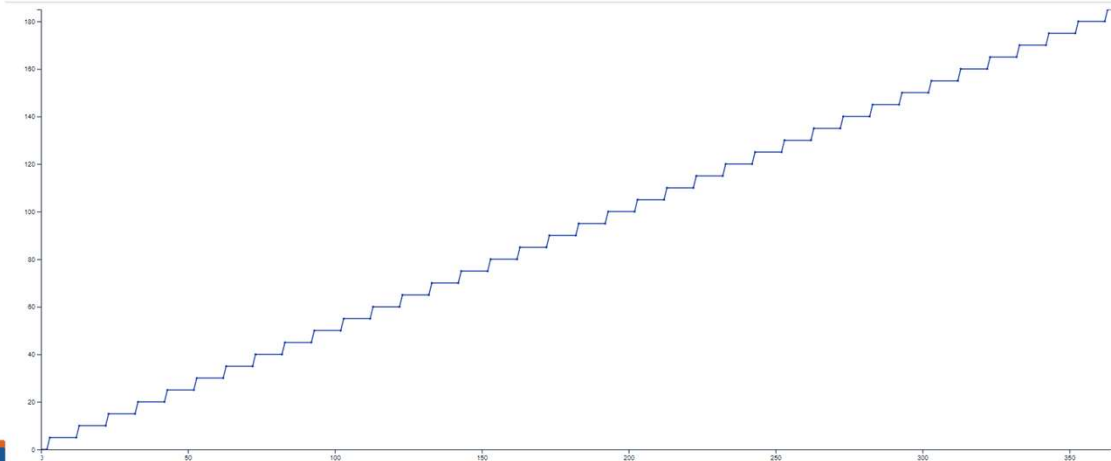
Scenario comparison: replenishment

- And we see that our replacement rate is behaving adequately.

FULFILLMENT (LATE ORDERS), FULFILLMENT RECEIVED (ORDERS ON-TIME)

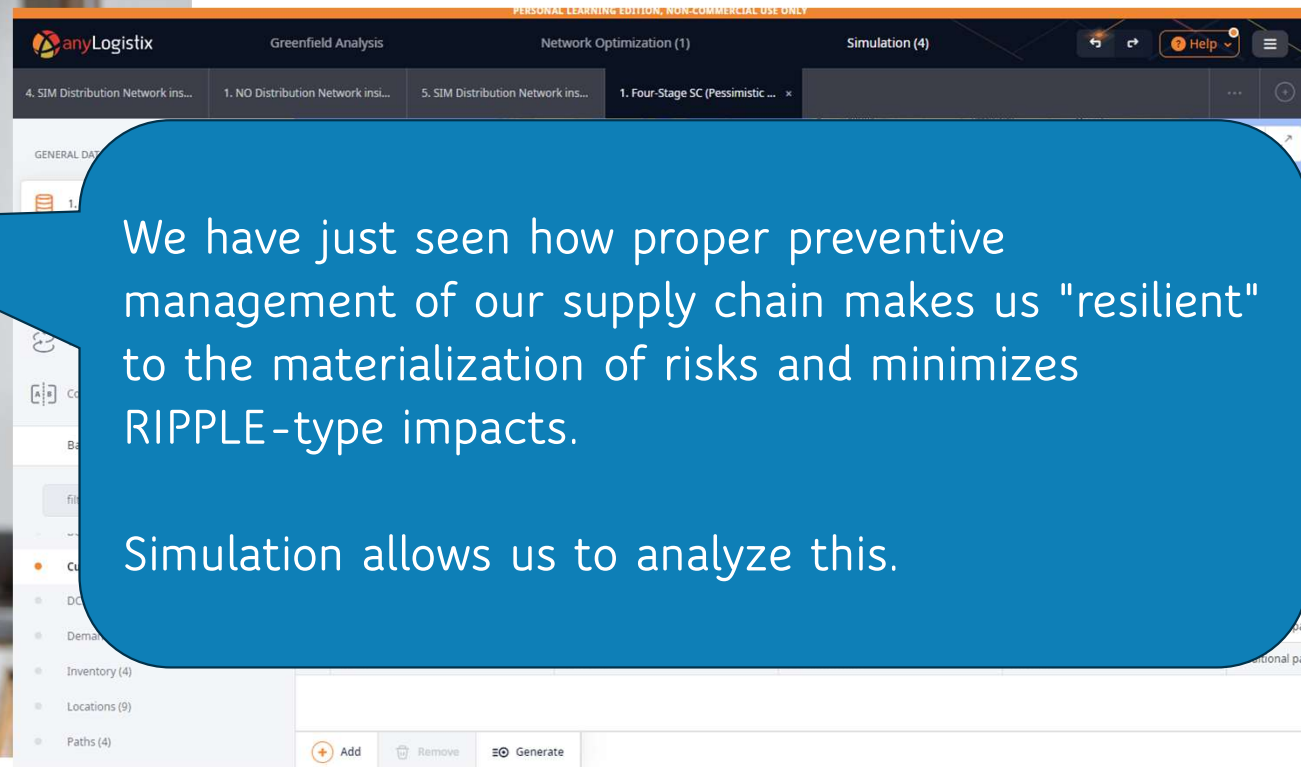


FULFILLMENT (LATE ORDERS), FULFILLMENT RECEIVED (ORDERS ON-TIME)





Prevention vs Reaction vs cost



Supplier Taiwan, Supplier China



Factory



DC



(All customers)





Learning recap

1. Develop analytical and management skills to analyze uncertainty in the supply chain.
2. We able to simulate uncertainty.
3. Two proactive strategies:
 - a) Increase inventory
 - b) Redundancy.